REPORT OF SCHEMA AND RELATIONSHIPS

(Data Analyst Internship)

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Language:	SQL

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INTRODUCTION

The **Finance Management System** project is developed to streamline and digitize the management of personal and organizational financial activities. In today's digital world, manual handling of financial data can lead to errors, inefficiency, and security concerns. This system provides a structured and secure way to manage essential financial components such as **user accounts, transactions, loans, payments, and investments** within a centralized platform.

This project aims to build a robust and user-friendly system that allows users to view and track their financial records, make transactions, manage loans and investments, and ensure timely payments. The admin module enables authorized personnel to monitor overall operations, manage user accounts, and maintain the integrity of financial data.

The system is designed using a **relational database model** to maintain data integrity and consistency, with well-defined relationships between entities such as users, accounts, transactions, loans, payments, and investments. It offers a clear interface for interaction, automated record-keeping, and the ability to generate accurate financial reports.

Overall, the Finance Management System enhances transparency, promotes efficient financial planning, and minimizes human error, making it a valuable tool for both personal and professional financial management.

SCHEMA OVERVIEW

The Finance Management System consists of seven interrelated tables that store and manage financial data for users, accounts, transactions, loans, payments, investments, and admins. The schema follows a relational structure to ensure data consistency, integrity, and easy access.

- **Database Name**: financemanagementsystem
- **Number of Tables**: 7
- **List of Tables**:
  - users
  - accounts
  - admins
  - investments
  - loans
  - payments
  - transactions

TABLES

- **USERS**

It is used to store and manage detailed information about each person who uses the system—whether for creating accounts, taking loans, making transactions, or investing money.

Column Name	Data Type	Constraints
user_id	INT	PRIMARY KEY, NOT NULL
full_name	VARCHAR(100)	NULL
email	VARCHAR(100)	UNIQUE, NULL
phone_number	VARCHAR(15)	NULL
address	TEXT	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

- **ACOUNTS**

It is responsible for storing and managing details about each financial account owned by users. It plays a vital role in tracking balances, account types, and linking transactions to specific users.

Column Name	Data Type	Constraints
account_id	INT	PRIMARY KEY, NOT NULL
user_id	INT	NULL
account_type	VARCHAR(50)	NULL
balance	DECIMAL(12,2)	NULL
opened_date	DATE	NULL
status	VARCHAR(20)	NULL

- **ADMINS**

It manages all system administrators who are responsible for overseeing user activity, managing records, and maintaining the system.

Column Name	Data Type	Constraints
admin_id	INT	PRIMARY KEY, NOT NULL
username	VARCHAR(50)	UNIQUE, NULL
password_hash	VARCHAR(255)	NULL
role	VARCHAR(50)	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

- **INVESTMENTS**

Tracks and manages user investments in financial products like mutual funds, fixed deposits, stocks, etc.

Column Name	Data Type	Constraints
investment_id	INT	PRIMARY KEY, NOT NULL
user_id	INT	NULL
investment_type	VARCHAR(50)	NULL
amount_invested	DECIMAL(12,2)	NULL
date_invested	DATE	NULL
current_value	DECIMAL(12,2)	NULL

- **LOANS**

Stores all **loan records** applied for or approved by users.

Column Name	Data Type	Constraints
loan_id	INT	PRIMARY KEY, NOT NULL
user_id	INT	NULL
loan_type	VARCHAR(50)	NULL
amount	DECIMAL(12,2)	NULL
interest_rate	DECIMAL(5,2)	NULL
start_date	DATE	NULL
end_date	DATE	NULL
status	VARCHAR(20)	NULL

- **PAYMENTS**

Manages all loan repayments made by users.

Column Name	Data Type	Constraints
payment_id	INT	PRIMARY KEY, NOT NULL
loan_id	INT	NULLL
amount_paid	DECIMAL(12,2)	NULL
payment_date	DATE	NULL
payment_method	VARCHAR(50)	NULL

- **TRANSACTIONS**

Keeps a record of all **financial activities** (deposits, withdrawals, transfers) in each user's account

Column Name	Data Type	Constraints
transaction_id	INT	PRIMARY KEY, NOT NULL
account_id	INT	FOREIGN KEY REFERENCES accounts
transaction_type	VARCHAR(50)	NULL
amount	DECIMAL(12,2)	NULL
transaction_date	DATE	NULL
description	TEXT	NULL

RELATIONSHIPS

Think of the users table as the **main person** in the system, and all other tables store the things that person does (like opening an account, making payments, taking a loan, etc.).

1. Users

This is the **main table**. Each user can have **multiple accounts ,** take **multiple loans** , make **multiple investments**

So users is **linked to**:

- **accounts** through **user_id**
- **loans** through **user_id**
- **investments** through **user_id**

2. Accounts

Each account **belongs to one user** and can have **many transactions**

So accounts is:

- Linked **to one user** (user_id)
- Linked **to many transactions** through **account_id**

3. Transactions

Each transaction is made **in one account**

So it is:

- Connected to the accounts table via **account_id**

4. Loans

Each loan **belongs to one user** and can have **many repayments** (payments)

So loans is:

- Connected **to one user** (user_id)
- Connected **to many payments** through loan_id

5. Payments

Each payment is made **toward one loan**

So it's:

- Linked to **loans** via **loan_id**

6. Investments

Each investment is made **by one user**

So it's:

- Connected to **users** via **user_id**

7. Admins

Admins are not connected to user transactions directly. They just **manage the system** (e.g., view, add, or update data)

In Short:

- **Users → Accounts → Transactions**
- **Users → Loans → Payments**
- **Users → Investments**
- **Admins** manage everything, but don't link to other tables

Conclusion

The Finance Management System is a practical and efficient solution for managing a wide range of financial activities in a structured and user-friendly way. This project brings together key financial operations such as user account management, transactions, loans, payments, and investments under a single system, allowing both users and administrators to perform their tasks with ease and accuracy.

With the help of a relational database, the system maintains proper connections between different entities, ensuring data consistency and smooth flow of information. Users can securely access their financial data, track account balances, monitor loan repayments, and manage investments, while administrators can oversee the system, manage user records, and generate useful insights.

This project not only demonstrates the application of database design and system logic but also highlights the importance of digital finance tools in today's world. It helps reduce manual errors, increases transparency, and supports better financial decision-making. Overall, the Finance Management System serves as a strong foundation for understanding real-world financial software and its impact on efficient money management.

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End of Report